Hw 4 - start from the lecture equation

1. What is u? [Lecture 1, slides 15-16]

The professor starts with the vacuum electric-field equation:

$$\nabla^2 E - \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} = 0$$

Take propagation along x and a transverse component E_y.

Write u(x,t)=E_y(x,t). Here c is the speed of light.

There is no y or z dependence, so the scalar equation is:

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0$$

2. Split the field into spatial waves. [Lecture 1, slide 16]

$$u(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} a(k,t)e^{ikx}\,dk$$

k is the wave number; a(k,t) is its amplitude at time t.

The integral adds all k. The prefactor is the Fourier convention.

3. Differentiate and substitute. [Lecture 1, slide 17]

$$\frac{\partial^2}{\partial x^2}e^{ikx} = (ik)^2 e^{ikx} = -k^2 e^{ikx}$$

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\left[-k^2 a - \frac{1}{c^2}\frac{\partial^2 a}{\partial t^2}\right]e^{ikx}\,dk = 0$$

Independent Fourier waves have independent coefficients:

$$-k^2 a - \frac{1}{c^2}\frac{\partial^2 a}{\partial t^2} = 0 \quad \Rightarrow \quad \frac{\partial^2 a}{\partial t^2} + c^2 k^2 a = 0$$

Solve the modes and obtain Eq. 7.90

4. Solve in time. [Lecture 1, slide 18]

Each k gives a harmonic oscillator with frequency omega(k).

$$a(k,t) = C(k)e^{-i\omega(k)t} + D(k)e^{i\omega(k)t}$$

In vacuum omega=c|k|, since k includes both signs.

In a dispersive medium, its material equations give omega(k).

As in Jackson, take real omega(k) with omega(-k)=omega(k).

5. A real field relates C and D.

Conjugating the Fourier integral gives a(-k,t)=a*(k,t).

$$C(-k)e^{-i\omega t} + D(-k)e^{i\omega t} = C^*(k)e^{i\omega t} + D^*(k)e^{-i\omega t}$$

Match the coefficients of the same time exponentials:

$$C(-k) = D^*(k) \quad \Rightarrow \quad D(k) = C^*(-k)$$

Insert into u. Change k to -k in the second integral:

$$u = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} C(k)e^{ikx - i\omega t}dk + \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} C^*(k)e^{-ikx + i\omega t}dk$$

The second integral is the complex conjugate (c.c.) of the first.

6. Match the professor to Jackson. [Lecture 8, slide 9]

The professor calls C(k) his A(k). Jackson uses twice that amplitude:

$$A_{\text{Jackson}}(k) = 2A_{\text{lecture}}(k) = 2C(k)$$

Thus C=A/2, where A below is Jackson's amplitude:

$$u(x,t) = \frac{1}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} A(k)e^{ikx - i\omega(k)t}dk + c.c.$$

This is Eq. 7.90. The two amplitude conventions give the same u.

Initial conditions and the lecture example

7. Find A(k) from the two initial conditions.

Let f(x)=u(x,0), g(x)=partial u(x,0)/partial t.

A hat means spatial Fourier transform, with the same convention:

$$\hat{f}(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(x)e^{-ikx}dx \qquad \text{(similarly for } \hat{g}\text{)}$$

In Eq. 7.90, change k to -k in c.c., then set t=0:

$$\hat{f}(k)=\frac{A(k)+A^{*}(-k)}{2}$$

Differentiate Eq. 7.90 in time, then set t=0:

$$\hat{g}(k)=\frac{-i\omega(k)}{2}[A(k)-A^{*}(-k)]$$

Multiply the second equation by i/omega and add to the first:

$$A(k)=\hat{f}(k)+\frac{i}{\omega(k)}\hat{g}(k)$$

$$A(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-ikx}\left[u(x,0)+\frac{i}{\omega(k)}\frac{\partial u}{\partial t}(x,0)\right]dx$$

This is Eq. 7.91: arbitrary initial shape and initial rate are included.

8. Recover exactly the professor's formula.

Lecture 8, slide 9 assumes g(x)=0. Therefore:

$$A_{\text{Jackson}}=\hat{f}, \qquad A_{\text{lecture}}=\frac{\hat{f}}{2}=\frac{1}{2\sqrt{2\pi}}\int_{-\infty}^{\infty}u(x,0)e^{-ikx}dx$$

Lecture 8, slide 10: f(x)=cos(k_0 x) then gives

$$u(x,t)=\frac{1}{2}\cos(k_0x-\omega_0t)+\frac{1}{2}\cos(k_0x+\omega_0t)$$

At t=0 this is cos(k_0 x), and its initial time derivative is zero.

Sources: Lec 1, 15-18; Lec 8, 9-10; Lec 9, 2; Jackson, 7.90-7.91.